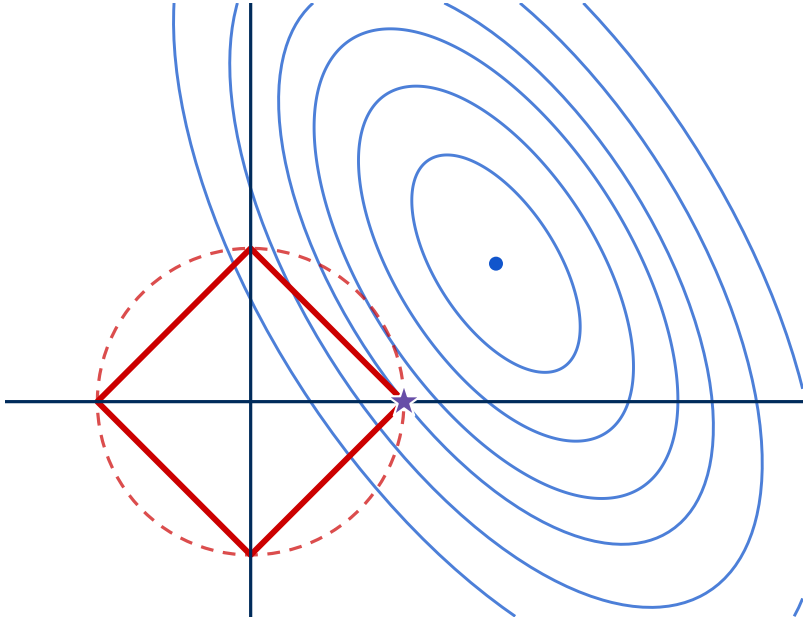


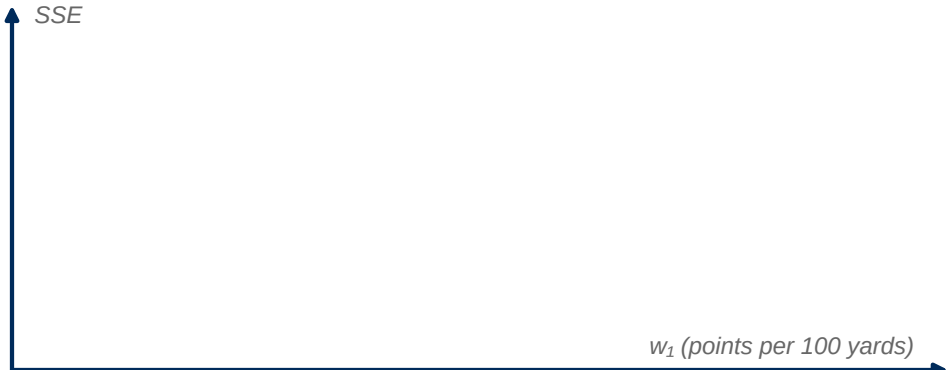
UNIT 02 · CHAPTER 3

Optimization and Regularization

CSE 30124 · Introduction to Artificial Intelligence
William Theisen · University of Notre Dame

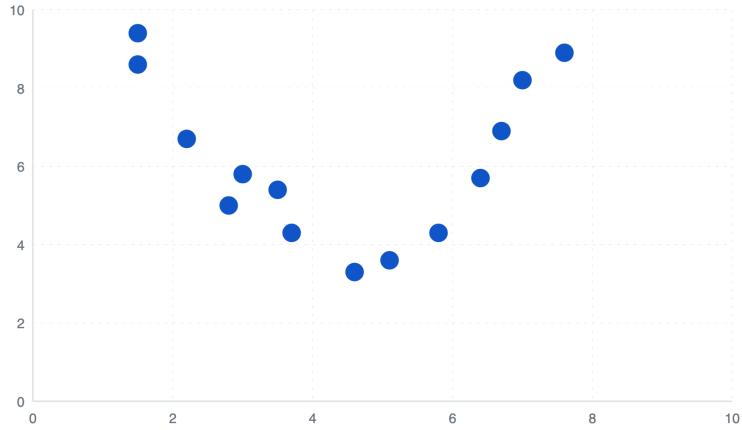
An iterative method

Instead of jumping straight to the bottom of the bowl, we can _____ a little at a time.



Overfitting

Unfortunately, **Linear Regressions** (a form of MLE), are **prone to overfitting**.



Regularization: L_1 and L_2

To help **prevent overfitting** when using **linear regressions**, we can perform a technique called **regularization**.

The goal of **regularization** is to **encourage models to learn the broader patterns** within the data rather than memorizing it.

L_2 regularization: Ridge

L2 regularization (called **ridge regularization**) **encourages** _____ **weights**.

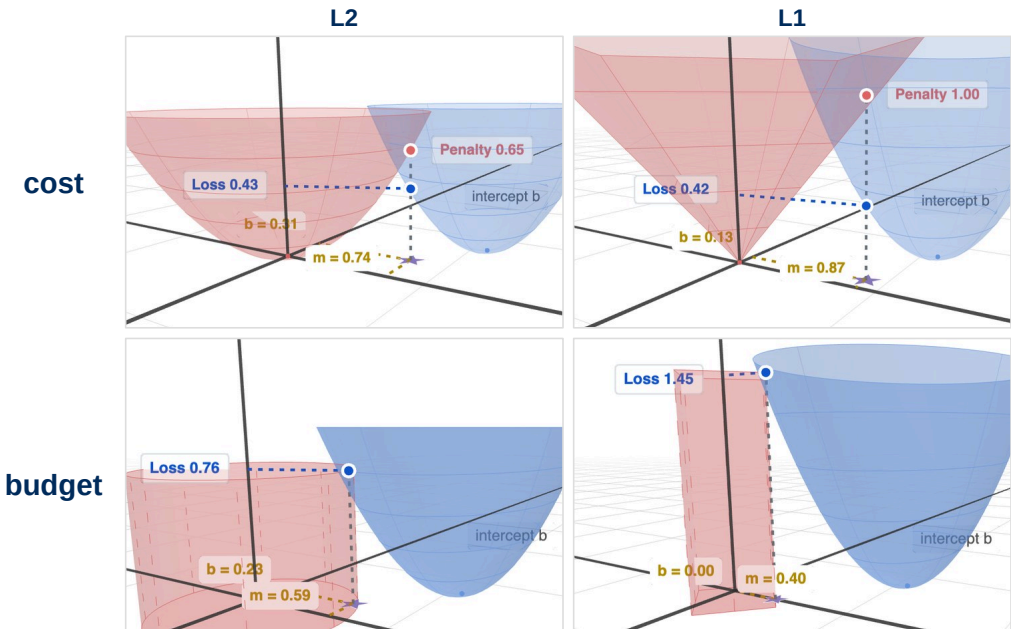
L_1 regularization: LASSO

L1 regularization (also called **LASSO**) leads to _____ and **sparse models**.

Formalizations of regularization

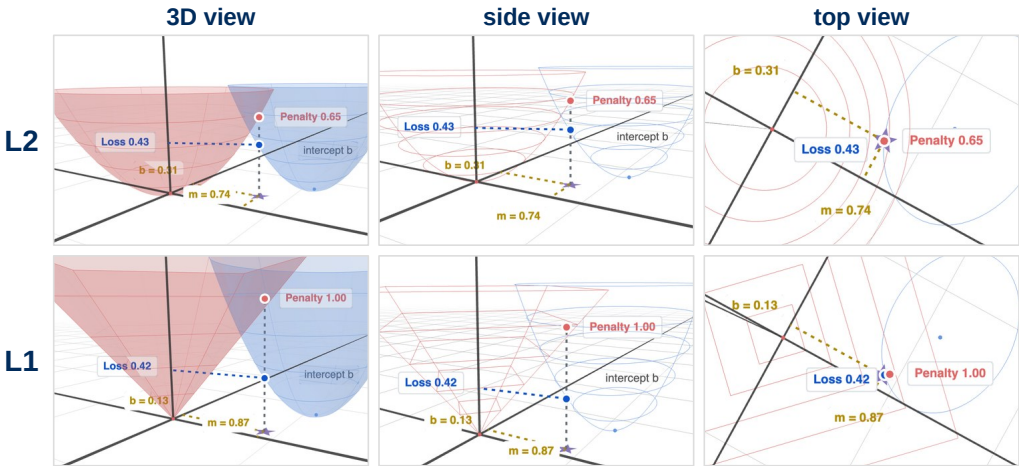
There are two ways to formalize regularization:

- As a _____ placed on the **weights**
- As a _____ for the **weights**



Regularization as a cost

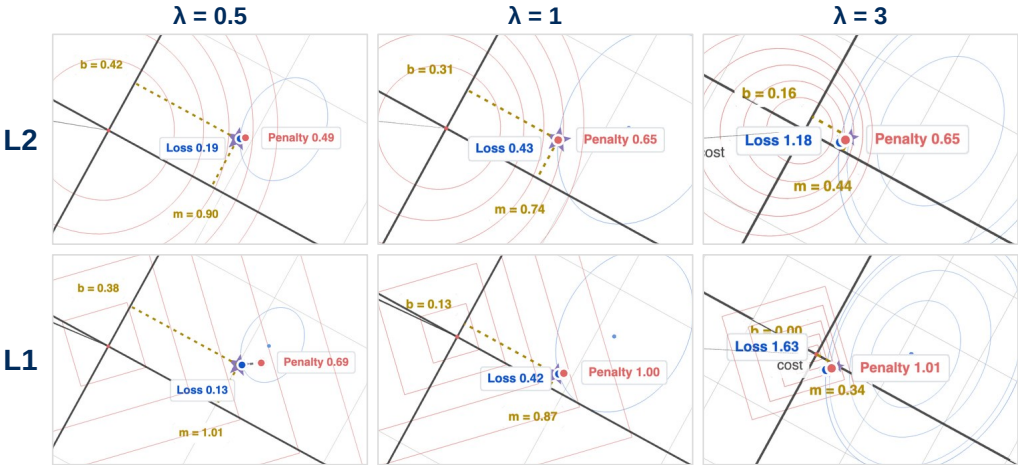
When we formalize **regularization** as a **cost**, we modify the **loss function** such that increasing the weights incurs a "cost".



We use an argument called λ to determine how heavily the **cost** is weighed. A larger λ increases the **cost**.

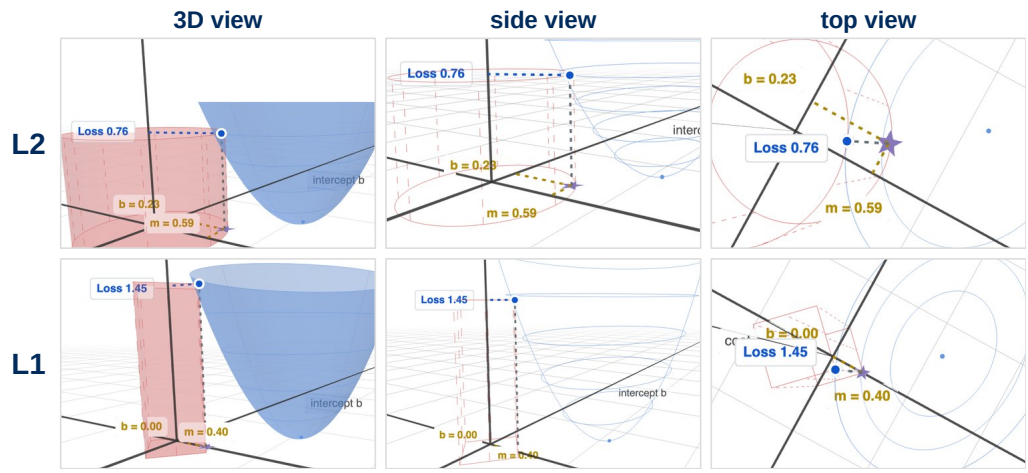
Regularization as a cost, increasing λ

A larger λ increases the **cost** of our weights, typically resulting in the optimum being pulled closer to the origin.



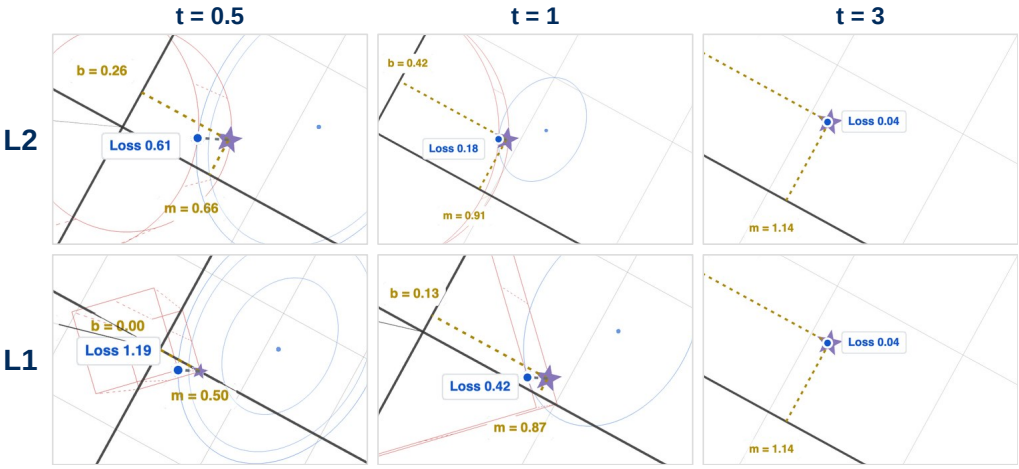
Regularization as a budget

When we formalize **regularization** as a **budget**, we constrain our **loss function** such that the weights chosen must remain within "budget".



Regularization as a budget, increasing t

A larger t increases the **budget** for our weights, typically resulting in the optimum being closer to the unconstrained optimum.



MAP estimation

L1 and **L2 regularization** are a form of **MAP Estimation**.

MAP stands for _____ which is a **Bayesian** approach for **applying a prior to our weights**.

Review

Complete the review: **L1 regularization** penalizes _____. **L2 regularization** penalizes _____. In a penalty formulation, increasing **lambda** makes the penalty _____. In a budget formulation, increasing the budget makes the constraint _____.

Matching: regularization

Write the letter of the matching definition next to each term.

___ **MAP Estimation**	**A.** Penalizes the squared size of the weights, encouraging small weights
___ **L2 (Ridge)**	**B.** Controls how heavily the regularization cost is weighed
___ **L1 (LASSO)**	**C.** The budget the weights must remain within
___ λ	**D.** Penalizes the absolute size of the weights, driving some to exactly 0
___ **t**	**E.** A Bayesian approach for applying a prior to our weights